Final proyect - Electrodynamics

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Activity Instructions

Ex. 1.0 **Obtain the wave equation and show that a plane wave is a solution.**

Ex. 2.0 **Obtain the relation for E_y and H_x for a plane wave such that $\vec{E} = (0, E_y, 0)$ and $\vec{H} = (H_x, 0, H_z)$ (TE polarization).**

Ex. 3.0 **Considering three media, propose three plane waves and apply the boundary conditions (Electrodynamics, D. Jackson, pages 16, 17 and 18). $k_x = 0$.**

Figura 1: Three-layer dielectric structure used in Problems 3 and 4.

Ex. 4.0 **Considering the attached figure, write the field in the three media with the corresponding boundary conditions.**

Ex. 5.0 **Using the values reported in Pochi Yeh (1988) (Fig. 11.3 and the values for TE waveguides, pages 304 and 307), plot**

the real part of the electric field depending on x and z.

Ex. 6.0 Make a 10 minute presentation (December 5th, starting at 6 pm).

Índice

Final Proyect

Problem 1: Wave equation in dielectrics

We start from Maxwell's equations in a **linear, homogeneous and source-free dielectric medium**, that is, with

$$\rho_f = 0, \qquad \vec{J}_f = 0$$

and the constitutive relations

$$\vec{D} = \varepsilon \vec{E}, \qquad \vec{B} = \mu \vec{H}$$

Thus, Maxwell's equations take the form:

$$\begin{aligned} \nabla \cdot \vec{D} &= 0, \\ \nabla \cdot \vec{B} &= 0, \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t}, \\ \nabla \times \vec{H} &= \frac{\partial \vec{D}}{\partial t} \end{aligned}$$

Using the constitutive relations, these become:

$$\begin{aligned} \nabla \cdot \vec{E} &= 0, \\ \nabla \times \vec{E} &= -\mu \frac{\partial \vec{H}}{\partial t}, \\ \nabla \times \vec{H} &= \varepsilon \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

The last two equations describe the time–space coupling between the electric and magnetic fields inside the dielectric.

Derivation of the wave equation for the electric field

We take the curl of Faraday's law:

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t}(\nabla \times \vec{H}) \tag{1}$$

The equation (**??**) tells us that the spatial rotation of the electric field is directly linked to the time variation of the magnetic field.

Substituting Ampère–Maxwell's law,

$$\nabla \times \vec{H} = \varepsilon \frac{\partial \vec{E}}{\partial t},$$

we obtain:

$$\nabla \times (\nabla \times \vec{E}) = -\mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} \tag{2}$$

We now use the vector identity:

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

Since the medium is source–free, $\nabla \cdot \vec{E} = 0$, therefore:

$$-\nabla^2 \vec{E} = -\mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} \tag{3}$$

Finally, we arrive at the **wave equation for the electric field**:

$$\boxed{\nabla^2 \vec{E} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0} \tag{4}$$

The equation (**??**) states that the electric field propagates as a wave inside the dielectric medium with a speed determined by the material parameters.

Wave equation for the magnetic field

Repeating the same procedure but starting from Ampère–Maxwell's law and taking the curl of $\nabla \times \vec{H}$, we obtain in an analogous way:

$$\boxed{\nabla^2 \vec{H} - \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0} \tag{5}$$

The equations (**??**) and (**??**) describe the propagation of electromagnetic waves in dielectrics.

Plane wave as a solution

We now assume a plane wave solution of the form:

$$\vec{E}(\vec{r}, t) = \vec{E}_0 \, e^{i(\vec{k}\cdot\vec{r} - \omega t)} \tag{6}$$

The expression (**??**) represents a wave of constant amplitude propagating in the direction of $\vec{k}$ with angular frequency ω.

Substituting this form into the wave equation (**??**), the spatial and temporal derivatives give:

$$\nabla^2 \vec{E} = -k^2 \vec{E}, \qquad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

Thus, the wave equation reduces to:

$$(-k^2 + \mu\varepsilon\omega^2)\, \vec{E} = 0 \tag{7}$$

For non–trivial solutions, we must have:

$$\boxed{k^2 = \mu\varepsilon\, \omega^2} \tag{8}$$

The relation (**??**) is the **dispersion relation** of electromagnetic waves in a dielectric medium.

Therefore, we conclude that:

$$\boxed{\text{A plane wave is indeed a solution of the electromagnetic wave equation in dielectrics.}} \tag{9}$$

Problem 2: TE polarization and relation between E_y and H_x

In this section we analyze a **plane electromagnetic wave with TE polarization**. The term **TE (Transverse Electric) polarization** means that the electric field has no component along the direction of propagation and is entirely perpendicular to the plane of incidence. Since the wave propagates in the x–z plane, the electric field must be oriented along the y direction, while the magnetic field lies in the x–z plane.

Accordingly, the electric and magnetic fields take the form:

$$\vec{E} = (0, E_y, 0), \qquad \vec{H} = (H_x, 0, H_z) \tag{10}$$

The expression (**??**) shows that the electric field is purely transverse along the y direction, whereas the magnetic field has components along x and z. We assume harmonic time dependence and propagation in the x–z plane.

Plane wave representation of the TE field

We now assume a plane wave solution for the electric field component:

$$E_y(x, z, t) = E_{y0}\, e^{i(k_x x + k_z z - \omega t)} \tag{11}$$

The equation (**??**) describes a monochromatic plane wave propagating in the direction defined by the wave vector

$$\vec{k} = (k_x, 0, k_z)$$

with angular frequency ω.

Relation between E_y and H_x

To obtain the relation between E_y and H_x, we start from Faraday's law:

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \tag{12}$$

From the field configuration (**??**), the x-component of the curl is:

$$(\nabla \times \vec{E})_x = \frac{\partial E_y}{\partial z} \tag{13}$$

The equation (**??**) tells us that the spatial variation of the transverse electric field along the z direction generates the magnetic component H_x.

Using the plane wave form (**??**), we compute:

$$\frac{\partial E_y}{\partial z} = i k_z E_y \tag{14}$$

On the other hand, the time derivative of the magnetic field gives:

$$-\mu \frac{\partial H_x}{\partial t} = i\omega\mu H_x \tag{15}$$

Substituting (**??**) and (**??**) into Faraday's law (**??**), we obtain:

$$i k_z E_y = i\omega\mu H_x \tag{16}$$

From this expression, the **fundamental relation between E_y and H_x for a TE plane wave** follows directly:

$$\boxed{H_x = \frac{k_z}{\omega\mu} E_y} \tag{17}$$

The relation (**??**) shows that the magnetic field component H_x is directly proportional to the spatial derivative of the transverse electric field along the direction of propagation.

Complementary relation for H_z

For completeness, we now use the z-component of Faraday's law:

$$(\nabla \times \vec{E})_z = -\frac{\partial E_y}{\partial x} \tag{18}$$

Using again the plane wave form (**??**), we compute:

$$\frac{\partial E_y}{\partial x} = ik_x E_y \tag{19}$$

Substitution into Faraday's law yields:

$$-ik_x E_y = i\omega\mu H_z \tag{20}$$

Hence, the second magnetic component is given by:

$$\boxed{H_z = -\frac{k_x}{\omega\mu} E_y} \tag{21}$$

Final TE field relations

From equations (**??**) and (**??**), the complete magnetic field associated with a TE plane wave can be written as:

$$\boxed{\vec{H} = \left(\frac{k_z}{\omega\mu} E_y,\, 0,\, -\frac{k_x}{\omega\mu} E_y \right)} \tag{22}$$

These expressions complete the requirement of **Problem 2**, showing explicitly the relation between the transverse electric field E_y and the magnetic components H_x and H_z for a plane electromagnetic wave with TE polarization.

Problem 3: Three-layer dielectric structure and boundary conditions ($k_x = 0$)

We now consider a **three-layer dielectric structure** as depicted in the Fig. **??**: two semi-infinite outer media with refractive index n_1 and a finite slab of thickness d with refractive index n_2 in the region $0 < z < d$. The planar interfaces are located at

$$z = 0, \qquad z = d,$$

and the unit normal to the interfaces is

$$\hat{n} = \hat{z},$$

pointing from medium I to medium II and from medium II to medium III.

We assume a TE-polarized plane wave incident normally on the structure, i.e.

$$k_x = 0,$$

so that the wave propagates purely along the z–direction. For TE polarization with $k_x = 0$ the fields can be written as

$$\vec{E} = (0, E_y, 0), \qquad \vec{H} = (H_x, 0, 0),$$

and from **Problem 2** we have the relation

$$\boxed{H_x = \frac{k_z}{\omega\mu} E_y, \qquad H_z = 0 \quad (k_x = 0)} \tag{23}$$

Plane waves in each medium

Let μ_1, ε_1 be the permeability and permittivity in media I and III, and μ_2, ε_2 those in medium II. For normal incidence ($k_x = 0$), the longitudinal components of the wave vector are

$$\begin{aligned} k_{1z} &= \omega\sqrt{\mu_1\varepsilon_1}, \\ k_{2z} &= \omega\sqrt{\mu_2\varepsilon_2}. \end{aligned} \tag{24}$$

Assuming harmonic time dependence $e^{-i\omega t}$, we keep only the spatial dependence. The tangential electric field (for TE, this is E_y) in each region is written as a superposition of forward and backward propagating plane waves:

- ⋆ **Region I** ($z < 0$, index n_1):

$$E_y^{(I)}(z) = A\, e^{ik_{1z}z} + B\, e^{-ik_{1z}z} \tag{25}$$

- ⋆ **Region II** ($0 < z < d$, index n_2):

$$E_y^{(II)}(z) = C\, e^{ik_{2z}z} + D\, e^{-ik_{2z}z} \tag{26}$$

- ⋆ **Region III** ($z > d$, index n_1):

$$E_y^{(III)}(z) = F\, e^{ik_{1z}z} \tag{27}$$

In (**??**) we have kept only the transmitted wave in region III (no wave incident from $z \to +\infty$), corresponding to an incident wave coming from region I.

From (**??**), the magnetic fields follow as

$$\boxed{H_x^{(j)}(z) = \frac{k_{jz}}{\omega\mu_j} E_y^{(j)}(z), \qquad j = I, II, III} \tag{28}$$

Boundary conditions from Jackson (Sect. I.5)

According to Jackson (Eqs. (1.17)–(1.20)), the boundary conditions at an interface between two media with surface charge density σ and surface current density $\vec{K}$ are:

$$\begin{aligned} (\vec{D}_2 - \vec{D}_1)\cdot\hat{n} &= \sigma, \\ (\vec{B}_2 - \vec{B}_1)\cdot\hat{n} &= 0, \\ \hat{n}\times(\vec{E}_2 - \vec{E}_1) &= 0, \\ \hat{n}\times(\vec{H}_2 - \vec{H}_1) &= \vec{K}. \end{aligned}$$

In our problem we assume $\sigma = 0$ and $\vec{K} = 0$, so the normal components of $\vec{D}$ and $\vec{B}$ and the tangential components of $\vec{E}$ and $\vec{H}$ are continuous across each interface.

For a planar interface with normal $\hat{n} = \hat{z}$, the tangential directions are x and y. Since our TE fields have E_y and H_x tangential to the interface, the boundary conditions reduce to:

$$E_y^{(I)} = E_y^{(II)}, \qquad H_x^{(I)} = H_x^{(II)} \quad \text{at } z = 0,$$

and

$$E_y^{(II)} = E_y^{(III)}, \qquad H_x^{(II)} = H_x^{(III)} \quad \text{at } z = d.$$

Using (**??**)–(**??**) and (**??**), the conditions at $z = 0$ become

$$\begin{aligned} E_y^{(I)}(0) &= E_y^{(II)}(0), \\ H_x^{(I)}(0) &= H_x^{(II)}(0), \end{aligned} \tag{29}$$

that is,

$$\boxed{A + B = C + D} \tag{30}$$

$$\boxed{\frac{k_{1z}}{\omega\mu_1}(A + B) = \frac{k_{2z}}{\omega\mu_2}(C + D)} \tag{31}$$

Similarly, the boundary conditions at $z = d$ read

$$\begin{aligned} E_y^{(II)}(d) &= E_y^{(III)}(d), \\ H_x^{(II)}(d) &= H_x^{(III)}(d), \end{aligned} \tag{32}$$

which explicitly give

$$\boxed{C\, e^{ik_{2z}d} + D\, e^{-ik_{2z}d} = F\, e^{ik_{1z}d}} \tag{33}$$

$$\boxed{\frac{k_{2z}}{\omega\mu_2}\left(C\, e^{ik_{2z}d} + D\, e^{-ik_{2z}d}\right) = \frac{k_{1z}}{\omega\mu_1} F\, e^{ik_{1z}d}} \tag{34}$$

The set of equations (**??**), (**??**), (**??**) and (**??**) is the **system of boundary conditions** that determines the reflection, transmission and internal field amplitudes in the three-layer dielectric structure for a TE wave with $k_x = 0$.

Summary of the three-wave construction

We can summarize the construction for **Exercise 3** as follows:

- We considered three media with refractive indices n_1, n_2, n_1, arranged along the z–axis with interfaces at $z = 0$ and $z = d$.
- For TE polarization with $k_x = 0$, the fields have the form $\vec{E} = (0, E_y, 0)$ and $\vec{H} = (H_x, 0, 0)$, with the relation (**??**).
- In each region, the tangential electric field is written as a superposition of plane waves, (**??**)–(**??**), and the magnetic field follows from (**??**).
- Using Jackson's boundary conditions (Eqs. (1.17)–(1.20) with $\sigma = 0$ and $\vec{K} = 0$), we obtained the continuity relations at $z = 0$ and $z = d$, summarized in (**??**)–(**??**).

These results complete the requirement of **Problem 3**, providing the explicit plane-wave forms in each medium and the corresponding boundary conditions for the case $k_x = 0$ in direct connection with Jackson's treatment on pages 16–18.

Problem 4: Fields in the three media from the given figure

We now use the specific field configuration shown in Fig. **??**. The structure is again a **three-layer slab**: medium I for $z < 0$, medium II for $0 < z < d$, and medium III for $z > d$, with refractive indices

$$n_1, \quad n_2, \quad n_1,$$

respectively. We assume harmonic time dependence $e^{-i\omega t}$ and write only the spatial part of the tangential electric field component E_x.

The figure prescribes the following **plane-wave ansatz** in each region:

$$E_{x1}(z) = A\, e^{-ik_{z1}z}, \qquad z < 0, \tag{35}$$

$$E_{x2}(z) = B\, e^{-ik_{z2}z} + C\, e^{+ik_{z2}z}, \qquad 0 < z < d, \tag{36}$$

$$E_{x3}(z) = D\, e^{+ik_{z1}z}, \qquad z > d. \tag{37}$$

The expressions (**??**)–(**??**) represent, respectively, an incident wave and a reflected wave in medium I (encoded in A and B inside medium II), a superposition of forward and backward waves inside the slab (coefficients B and C), and a transmitted wave propagating in medium III (coefficient D).

Boundary conditions for E_x and the associated magnetic field

According to Jackson's boundary conditions (Sect. I.5), if there is no surface charge density σ and no surface current density $\vec{K}$ at the interfaces, then:

- the **tangential component of $\vec{E}$** is continuous across the interface,
- the **tangential component of $\vec{H}$** is also continuous.

For our planar interfaces with normal $\hat{n} = \hat{z}$, the component E_x is tangential, and the relevant magnetic component is H_y. From Maxwell's equations for TE-like waves we can write, in each medium,

$$H_y^{(j)}(z) = \frac{1}{i\omega\mu_j}\frac{dE_{xj}(z)}{dz}, \qquad j = 1{,}2{,}3, \tag{38}$$

so that continuity of H_y translates into continuity of $\frac{1}{\mu}\frac{dE_x}{dz}$.

Let us compute the derivatives of the fields defined in (**??**)–(**??**):

$$\frac{dE_{x1}}{dz} = -ik_{z1}A\, e^{-ik_{z1}z}, \tag{39}$$

$$\frac{dE_{x2}}{dz} = -ik_{z2}B\, e^{-ik_{z2}z} + ik_{z2}C\, e^{+ik_{z2}z}, \tag{40}$$

$$\frac{dE_{x3}}{dz} = +ik_{z1}D\, e^{+ik_{z1}z}. \tag{41}$$

Boundary conditions at $z = 0$

At the first interface ($z = 0$) we impose:

- Continuity of E_x:

$$E_{x1}(0) = E_{x2}(0), \tag{42}$$

  that is,

$$\boxed{A = B + C} \tag{43}$$

- Continuity of H_y, equivalently of $\frac{1}{\mu}\frac{dE_x}{dz}$:

$$\frac{1}{\mu_1}\left.\frac{dE_{x1}}{dz}\right|_{z=0} = \frac{1}{\mu_2}\left.\frac{dE_{x2}}{dz}\right|_{z=0}, \tag{44}$$

  which yields

$$\boxed{-\frac{ik_{z1}}{\mu_1}A = -\frac{ik_{z2}}{\mu_2}B + \frac{ik_{z2}}{\mu_2}C} \tag{45}$$

The relations (**??**) and (**??**) are the boundary conditions connecting the amplitudes A,B,C at $z = 0$.

Boundary conditions at $z = d$

At the second interface ($z = d$) between media II and III, we again impose continuity of E_x and H_y:

- Continuity of E_x:

$$E_{x2}(d) = E_{x3}(d), \tag{46}$$

  that is,

$$\boxed{B\, e^{-ik_{z2}d} + C\, e^{+ik_{z2}d} = D\, e^{+ik_{z1}d}} \tag{47}$$

- Continuity of H_y (or $\frac{1}{\mu}\frac{dE_x}{dz}$):

$$\frac{1}{\mu_2}\left.\frac{dE_{x2}}{dz}\right|_{z=d} = \frac{1}{\mu_1}\left.\frac{dE_{x3}}{dz}\right|_{z=d}, \tag{48}$$

  which gives

$$\boxed{-\frac{ik_{z2}}{\mu_2}B\, e^{-ik_{z2}d} + \frac{ik_{z2}}{\mu_2}C\, e^{+ik_{z2}d} = \frac{ik_{z1}}{\mu_1}D\, e^{+ik_{z1}d}} \tag{49}$$

Final system for the amplitudes

Collecting (**??**), (**??**), (**??**) and (**??**), we obtain the complete **set of boundary conditions** relating the four amplitudes A,B,C,D:

$$\boxed{\begin{aligned} A &= B + C, \\ -\frac{k_{z1}}{\mu_1}A &= -\frac{k_{z2}}{\mu_2}B + \frac{k_{z2}}{\mu_2}C, \\ B\, e^{-ik_{z2}d} + C\, e^{+ik_{z2}d} &= D\, e^{+ik_{z1}d}, \\ -\frac{k_{z2}}{\mu_2}B\, e^{-ik_{z2}d} + \frac{k_{z2}}{\mu_2}C\, e^{+ik_{z2}d} &= \frac{k_{z1}}{\mu_1}D\, e^{+ik_{z1}d}. \end{aligned}} \tag{50}$$

This system, together with the prescribed incident amplitude (for example fixing A), allows us to solve for B, C and D. Thus we have written the fields in the three media exactly as in the given figure and derived the corresponding boundary conditions required in **Exercise 4**.

Problem 5: TE slab modes and field profiles (Pochi Yeh, 1988)

In this problem we follow the treatment of Pochi Yeh (1988) for a **symmetric slab waveguide** with refractive indices

$$n_2 = 2{,}5, \qquad n_1 = 1{,}5$$

where n_2 corresponds to the guiding core and n_1 to the cladding layers. A *slab waveguide* is a planar dielectric structure in which a thin high–index core layer of index n_2 is sandwiched between two semi–infinite lower–index claddings of index n_1; the structure is infinite in the $y-$ and z–directions and has a finite thickness only along x. Light is confined in the transverse x–direction by total internal reflection at the two planar interfaces and propagates along z.

The core thickness is taken as

$$d = \lambda$$

that is, equal to one free–space wavelength.

We consider **guided TE modes** in this structure. As in Yeh, the electric field has only the y-component, while the magnetic field has $x-$ and z–components:

$$\vec{E} = (0, E_y, 0), \qquad \vec{H} = (H_x, 0, H_z)$$

TE modal form and transverse profile

For each guided TE mode, the electric field is written as

$$E_y(x,z,t) = E_m(x)\, e^{i(\omega t - \beta z)} \tag{51}$$

The function $E_m(x)$ describes the **transverse mode profile** and depends only on the coordinate x measured across the slab. Inside the slab ($|x| < \frac{d}{2}$) the field oscillates, while in the cladding ($|x| > \frac{d}{2}$) it decays exponentially. Following Yeh, we write this piecewise as

$$E_m(x) = \begin{cases} A\,\sin(hx) + B\,\cos(hx), & |x| < \dfrac{d}{2} \\ C\,\exp(-qx), & x > \dfrac{d}{2} \\ D\,\exp(+qx), & x < -\dfrac{d}{2} \end{cases} \tag{52}$$

The equation (**??**) tells us that the field behaves like a standing wave in the core and decays away from the core into the cladding.

The parameters h and q are related to the propagation constant β by

$$\boxed{h = k_0\sqrt{n_2^2 - n_{\text{eff}}^2}, \qquad q = k_0\sqrt{n_{\text{eff}}^2 - n_1^2}} \tag{53}$$

where

$$k_0 = \frac{\omega}{c}, \qquad n_{\text{eff}} = \frac{\beta}{k_0}$$

is the **effective index of refraction of the mode** (normalized propagation constant).

Eigenvalue conditions and symmetry of the modes

At the interfaces $x = \pm\frac{d}{2}$ the tangential components of $\vec{E}$ and $\vec{H}$ must be continuous. For TE modes, this implies the continuity of E_y and H_z, which in turn requires the continuity of $E_m(x)$ and its derivative $\partial E_m/\partial x$ at $|x| = \frac{d}{2}$.

Imposing these conditions on (**??**) leads to two families of solutions, which Yeh classifies as:

- **Even TE modes** (symmetric with respect to $x = 0$):

$$E_m(-x) = E_m(x)$$

In this case, the transverse field in the core can be written purely as a cosine:

$$\boxed{E_m^{(\text{even})}(x) = \begin{cases} B\cos(hx), & |x| < \dfrac{d}{2} \\ B\cos\left(\dfrac{hd}{2}\right) e^{-q(|x|-d/2)}, & |x| > \dfrac{d}{2} \end{cases}} \tag{54}$$

and the boundary conditions yield the eigenvalue equation

$$\boxed{h\tan\left(\frac{hd}{2}\right) = q} \tag{55}$$

- **Odd TE modes** (antisymmetric with respect to $x = 0$):

  $$E_m(-x) = -E_m(x)$$

  Now the core field is purely sinusoidal:

  $$\boxed{E_m^{(\text{odd})}(x) = \begin{cases} A\sin(hx), & |x| < \dfrac{d}{2} \\ A\,\mathrm{sgn}(x)\sin\left(\dfrac{hd}{2}\right) e^{-q(|x|-d/2)}, & |x| > \dfrac{d}{2} \end{cases}} \tag{56}$$

  and the matching conditions give

  $$\boxed{h\cot\left(\frac{hd}{2}\right) = -q} \tag{57}$$

Equations (**??**) and (**??**) are exactly the eigenvalue relations discussed by Yeh for TE modes. They determine the allowed pairs (h, q) and therefore the discrete set of effective indices n_{eff} (guided modes).

It is convenient to introduce the **normalized frequency parameter**

$$\boxed{V^2 = (n_2^2 - n_1^2)\left(\frac{\pi d}{\lambda}\right)^2} \tag{58}$$

so that the allowed TE modes correspond to points (u, v) in the plane

$$u = \frac{hd}{2}, \qquad v = \frac{qd}{2}$$

satisfying

$$\boxed{u^2 + v^2 = V^2} \tag{59}$$

together with either the even condition $v = u\tan u$ or the odd condition $v = -u\cot u$ as shown graphically in Yeh's Fig. 11.2.

Numerical values for the example of Fig. 11.3

For the specific slab considered in Fig. 11.3, Yeh takes

$$n_2 = 2{,}5, \qquad n_1 = 1{,}5, \qquad d = \lambda$$

Using (**??**), the normalized frequency is

$$\begin{aligned} V^2 &= (n_2^2 - n_1^2)\left(\frac{\pi d}{\lambda}\right)^2 \\ &= (2{,}5^2 - 1{,}5^2)\pi^2 = (6{,}25 - 2{,}25)\pi^2 = 4\pi^2 \end{aligned} \tag{60}$$

so that

$$\boxed{V = 2\pi} \tag{61}$$

For this value of V, Yeh shows that there are **four guided TE modes** (labeled $m = 0,1,2,3$), with normalized propagation constants $\bar{\beta} = \beta/(\omega/c)$ given approximately by

$$\boxed{\bar{\beta}_0 = 2{,}463, \quad \bar{\beta}_1 = 2{,}348, \quad \bar{\beta}_2 = 2{,}148, \quad \bar{\beta}_3 = 1{,}850} \tag{62}$$

Each of these values lies between $n_1 = 1{,}5$ and $n_2 = 2{,}5$, as required for guided modes.

From (**??**), using $n_{\mathrm{eff},m} = \bar{\beta}_m$, the corresponding transverse parameters for each mode are

$$\boxed{h_m = k_0\sqrt{n_2^2 - \bar{\beta}_m^2}, \qquad q_m = k_0\sqrt{\bar{\beta}_m^2 - n_1^2}} \tag{63}$$

for $m = 0,1,2,3$.

The pairs (h_m, q_m) satisfy the circle condition (**??**) and either the even or odd eigenvalue equation:

- $m = 0$ and $m = 2$: even TE modes, described by (**??**) and (**??**)
- $m = 1$ and $m = 3$: odd TE modes, described by (**??**) and (**??**)

Thus, the transverse field profiles plotted in Fig. 11.3 are proportional to

$$E_m^{(\mathrm{even})}(x) \quad \text{or} \quad E_m^{(\mathrm{odd})}(x)$$

with the appropriate values of h_m and q_m given by (**??**).

Real part of the electric field as a function of x and z

Finally, for each mode m, the electric field can be written in the form (**??**) as

$$E_y^{(m)}(x,z,t) = E_{0m}\, E_m(x)\, e^{i(\omega t - \beta_m z)} \tag{64}$$

where E_{0m} is a (real) amplitude factor.

The **real part of the electric field**, which is the quantity we will plot in the next step of the project, is therefore

$$\boxed{\Re\big[E_y^{(m)}(x,z,t)\big] = E_{0m}\, E_m(x)\, \cos\big(\omega t - \beta_m z\big)} \tag{65}$$

Equation (**??**) tells us that all the spatial structure in x is encoded in the mode profile $E_m(x)$, while the dependence on the longitudinal coordinate z appears through the phase factor $\cos(\omega t - \beta_m z)$. In the following step we will use (**??**), (**??**), (**??**) and (**??**) to generate numerical plots of $\Re[E_y^{(m)}(x,z,t_0)]$ as a function of x and z for the TE modes of the symmetric slab waveguide described by Yeh.

TE_slab_modes_Yeh_high_resolution.png

Figura 2: **Distribución espacial del campo eléctrico real** $\Re[E_y(x,z)]$ para los cuatro primeros modos TE guiados en una guía de onda tipo slab simétrica con $n_2 = 2{,}5$, $n_1 = 1{,}5$ y $d = \lambda$. Las líneas verticales punteadas indican las fronteras del núcleo en $x = \pm d/2$. El gradiente de color representa la amplitud y el signo del campo: colores cálidos (rosa–naranja) corresponden a máximos positivos, mientras que los tonos oscuros (azul–negro) indican valores negativos. Cada subfigura corresponde a un modo TE $m = 0{,}1{,}2{,}3$, mostrando los patrones de interferencia longitudinal y la estructura transversal par/impar característica de cada modo.

Minitablita de parámetros del modelo de guía de onda slab

n_2	Índice de refracción del núcleo (región guiada), aquí $n_2 = 2{,}5$
n_1	Índice de refracción de los claddings (regiones de decaimiento evanescente), $n_1 = 1{,}5$
d	Espesor del núcleo de la guía de onda; en este modelo $d = \lambda$
β_m	Constante de propagación del modo m, que determina la variación longitudinal del campo como $\cos(\beta_m z)$
h_m	Parámetro transversal oscilatorio del campo dentro del núcleo $h_m = k_0\sqrt{n_2^2 - n_{\text{eff}}^2}$
q_m	Parámetro de decaimiento evanescente en los claddings $q_m = k_0\sqrt{n_{\text{eff}}^2 - n_1^2}$
E_y	Componente transversal del campo eléctrico para modos TE guiados
m	Índice modal que clasifica los modos guiados $m = 0{,}1{,}2{,}3$

Cuadro 1: Parámetros físicos y geométricos del modelo de guía de onda slab para los modos TE guiados simulados numéricamente.

La Fig. **??** muestra cómo el campo eléctrico queda *confinado* alrededor de la región central de la guía de onda debido a la reflexión total interna en las fronteras del material. Esto hace que la energía de la onda permanezca guiada principalmente dentro del núcleo. Al aumentar el índice modal m, aparecen más nodos en la dirección transversal x, lo cual indica que el patrón del campo se vuelve cada vez más complejo y corresponde a modos de orden superior.

Los modos pares son simétricos respecto al plano central $x = 0$, es decir, el campo tiene la misma forma a ambos lados del centro, mientras que los modos impares son antisimétricos, cambiando de signo al cruzar dicho plano. Finalmente, la variación del campo a lo largo de la dirección de propagación z está gobernada por el factor $\cos(\beta_m z)$, que describe la oscilación periódica del campo conforme la onda avanza dentro de la guía.